<!-- filename: ai_semiconductor_industry_report_2025.md -->

AI Semiconductor Market — 2025 Annual Review

*Published by **Meridian Technology Research** — January 15, 2026*

> Data note: Company-level figures for NexaCore and QuantumEdge are Meridian's calendar-year 2025 estimates, reconciled to each company's reported fiscal-year 2025 results. Meridian is a third-party research firm; its analysis is not a company filing.

Executive Summary

The AI semiconductor market reached **$142B in calendar year 2025**, up from **$68B in 2024 (+109%)**. **NexaCore Technologies** dominates with **$61.2B** in revenue, representing **43%** of the total market. The growth trajectory is expected to continue, with the market forecast at **$220–240B in 2026**.

(Source: Meridian Technology Research, AI Semiconductor Market 2025 Annual Review, Executive Summary)

Market Structure and Share

Company	CY2025 Revenue (est.)	Market Share
NexaCore Technologies	$61.2B	43%
QuantumEdge Corp	$28.3B	20%
AMD (AI segment est.)	$11.4B	8%
Google TPU (internal)	$9.8B est.	7%
Amazon Trainium (internal)	$7.2B est.	5%
Intel Gaudi	$4.1B	3%
Other	$19.4B	14%

Note that these are **total AI semiconductor market shares**, distinct from the **AI training accelerator share** (NexaCore ~72%, QuantumEdge ~18%) cited in company filings.

(Source: Meridian Technology Research, Market Structure and Share)

NexaCore Analysis

NexaCore's FY2025 results corroborate the picture of a market leader at scale: revenue of **$61.2B**, gross margin of **76%**, and net income of **$22.1B**. The **X3 ramp** is described as "unprecedented in scale."

NexaCore's **SilicaFab dependence** is characterized as "the single largest concentration risk in the semiconductor industry."

Meridian notes: "NexaCore's 72% training market share and 76% gross margin suggest pricing power that may attract antitrust scrutiny." (The US Department of Justice issued a civil investigative demand to NexaCore in October 2025.)

Financial Commentary

NexaCore's FY2025 gross margin of 76% significantly exceeds QuantumEdge's 65%, however Meridian analysis suggests NexaCore's true adjusted gross margin, excluding one-time capacity reservation fee amortization, may be closer to 71%.

Software and Mix

NexaOS DataBridge and X3 premium pricing contributed to margin expansion from 71.9% in FY2024.

(Source: Meridian Technology Research, NexaCore Analysis)

QuantumEdge Analysis

QuantumEdge reported revenue of **$28.3B (+51%)** and a gross margin of **65%**. **QuantumCloud** is growing rapidly, with **180,000 customers** and **$1.4B** in revenue.

Meridian rates QuantumEdge **"BUY"** with an FY2026 revenue target of **$39B**, within the **$38–42B** company guidance. Meridian describes the patent litigation against NexaCore as a **"binary risk"**: a favorable outcome could provide licensing leverage, while an adverse one could result in costs and distraction.

(Source: Meridian Technology Research, QuantumEdge Analysis)

Comparative Analysis

NexaCore vs QuantumEdge — FY2025

Metric	NexaCore	QuantumEdge	Commentary
Revenue	$61.2B	$28.3B	NexaCore 2.2× larger
Revenue Growth	+107%	+51%	NexaCore accelerating faster
Gross Margin	76.0%	65.0%	11 point gap, pricing power
Operating Margin	40.5%	20.8%	NexaCore nearly 2× more profitable
R&D % of Revenue	15.4%	17.0%	QuantumEdge investing more as % of rev
Net Income	$22.1B	$4.8B	NexaCore 4.6× more profitable
Free Cash Flow	$19.0B	$4.1B	NexaCore 4.6× more FCF
EPS	$8.84	$4.24	NexaCore 2.1× higher
Market Share (Training)	72%	18%	—
Taiwan Supply Risk	Yes (100%)	No	QuantumEdge advantage

(Source: Meridian Technology Research, Comparative Analysis)

Geopolitical Risk — Taiwan Semiconductor Concentration

The **geographic fabrication risk** posed by Taiwan is the central structural vulnerability of the AI chip market. NexaCore's dependence on SilicaFab is "unparalleled in the semiconductor industry." QuantumEdge's use of **GlobalChip in South Korea** "reduces but does not eliminate geopolitical supply risk."

Meridian estimates that a Taiwan Strait crisis would reduce AI training capacity additions by **65–75% industrywide for 12–18 months**.

(Source: Meridian Technology Research, Geopolitical Risk)

China Export Controls

US **EAR Part 744** restrictions effective Q2 2025 reduced combined industry China revenue by an estimated **$14.2B in CY2025**. NexaCore's China revenue fell from **$5.9B (FY2024)** to **$4.9B (FY2025)**. Further expansion of restrictions to cover **X3-class chips** is expected by mid-2026.

(Source: Meridian Technology Research, China Export Controls)

HBM Memory Bottleneck

Both NexaCore and QuantumEdge are competing for **HBM4** from **HB Memory Corp** and **ShinKwang Electronics**. Industry HBM4 capacity is limited to approximately **2.4B units per year** against estimated demand of **4.2B units**. This bottleneck will constrain AI chip production through 2026.

(Source: Meridian Technology Research, HBM Memory Bottleneck)

Outlook 2026

- Total market forecast: **$220–240B**.
- NexaCore X3 at scale plus X4 development means "no credible challenger to NexaCore's training dominance through at least 2027."
- QuantumEdge's **QE-Ultra** is the most significant competitive test; if successful, QuantumEdge could recover training share to **22–25%**.
- A rumored **IPO of SilicaFab** would change the manufacturing landscape.

(Source: Meridian Technology Research, Outlook 2026)